Restoring the Lost Gaps: An IRT Debiasing Framework for Silicon Sampling

A Research Memo for Hubert Cadieux & Justin Savoie on Subgroup Flattening

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1. Context & The Core Puzzle: Why Models Lost to a 2021 Poll

In our evaluation of Large Language Models simulating Canadian public opinion across 15 demographic subgroups (5 regions $\times$ 3 age cohorts) on 10 policy items from the 2025 Canadian Election Study ($N = 20,180$), two independent programs yielded clear empirical baselines:

1. **Justin Savoie (In-Context Prompting)**: Evaluated zero-shot hosted models (`Qwen3.5-9B` and `Qwen3.8-2.4T-A95B` via Together API) on 1,500 CES 2021 personas across Basic, Rich, and Partisan conditions. Best prompted model: **Qwen 9B Rich + Partisan (Mean Subgroup TV = 0.291; National TV = 0.258)**.
2. **Hubert Cadieux (Domain Fine-Tuning)**: Fine-tuned `Llama-3.3-70B-Instruct` on Azure AI Foundry on 8 Canadian surveys (2019–2024; 20k/50k/100k samples). Best fine-tuned model: **Fixed context, 50k examples (Mean Subgroup TV = 0.178; National TV = 0.157)**.

Despite fine-tuning roughly halving raw model error ($0.325 \rightarrow 0.178$), **both models were defeated on 8 of 10 items by an unchanged four-year-old survey (CES 2021, Subgroup TV = 0.151)**.

Cadieux's Error Decomposition

Cadieux identified the precise mechanism behind this defeat: - **National-level error**: Both the 2021 survey (0.141) and the fine-tuned 70B model (0.157) have comparable error regarding where Canada stands nationally. - **Subgroup difference error**: While CES 2021 captures **70% of actual subgroup gap magnitude** (correlation 0.71), the fine-tuned model captures only **31% of gap magnitude** (correlation 0.42). The models severely flatten demographic groups toward each other. - When Cadieux gave the CES 2021 subgroup gaps the real 2025 national figure, TV plummeted to **0.064**—almost reaching the human measurement noise floor (**0.062**).

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2. Mechanistic Diagnosis: The Discrimination Slope Deficit

To understand why LLMs flatten subgroups, we fitted an **Ordered-Probit Item Response Theory (IRT)** model to both human respondents in CES 2021 ($N = 19,468$) and simulated LLM personas ($N = 1,500$):

$$P(Y_{ij} \leq k \mid \theta_i) = \Phi(c_{j,k} - \alpha_j \theta_i)$$

where $\theta_i \sim \mathcal{N}(0,1)$ is latent political ideology (calibrated from Left-Right self-placement and party identification), α_j is item discrimination, and $c_{j,k}$ are difficulty thresholds.

Table 1 presents the fitted discrimination parameters across all policy items:

Policy Item	Human α_{2021}	LLM Basic α_{LLM}	LLM Partisan α_{LLM}	Flattening Ratio ($\alpha^{basic}_{LLM}/\alpha_{human}$)
`pos_energy` (Pipelines)	**+0.493**	+0.117	+0.431	**23.7%**
`pos_jobs` (Jobs vs. Env)	**+0.462**	+0.128	+0.477	**27.7%**
`spend_defence`	**+0.329**	+0.043	+0.483	**13.1%**
`imm` (Immigration level)	**−0.321**	−0.114	−0.386	**35.5%**
`spend_rec_indi` (Reconciliation)	**−0.351**	−0.128	−0.672	**36.4%**
`spend_nation_c` (Childcare)	**−0.325**	−0.124	−0.587	**38.1%**
`spend_afford_h` (Housing)	**−0.357**	−0.141	−0.475	**39.5%**
`pos_life` (MAID)	**−0.219**	−0.035	−0.515	**16.0%**

Key Mechanistic Findings

1. **The LLM gets every single sign/polarity right**: It correctly understands that conservatives and older cohorts lean toward pipelines, defence, and prioritizing jobs, while progressives and younger cohorts favor childcare, housing, and reconciliation.
2. **The slope is compressed by ~3x**: In the basic demographic condition, the model's discrimination slope is shrunk to **28%–39% of human magnitude** (mean $\approx 31\%$). This provides exact mathematical proof of Cadieux's observation: *the model knows which direction groups differ, but compresses the distance between them.*
3. **In-context partisan cues over-correct**: While prompting with party ID inflates α, it does so unevenly, causing extreme over-polarization on sensitive items like Indigenous reconciliation ($\alpha = -0.672$ vs human -0.351).

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3. The Architecture: Approach B (IRT Debiasing & Combiner)

Rather than asking an LLM to simultaneously predict national consensus and resolve demographic variance inside a single prompt, our framework **decouples the estimation into two specialized components**:

1. **Item Macro Level ($\vec{\tau}_j$)**: The LLM acts solely as a macro forecaster, providing national cumulative marginal distributions $\hat{F}^{nat}_{j,k}$. These map to baseline cutoffs:
$$\tau_{j,k} = \Phi^{-1}\left(\hat{F}^{nat}_{j,k}\right) \cdot \sqrt{\alpha_j^2 + 1}$$
2. **Demographic Trait Distributions ($p(\theta \mid g)$)**: Grounded in stable survey psychometrics. Subgroup means $\mu_g = \mathbb{E}[\theta \mid \text{Region, Age}]$ and variances $\sigma_g^2 = \text{Var}\ (\theta \mid g)$ are estimated from historical data (CES 2021). These cleavages remain structurally stable across election cycles.
3. **IRT Reintegration**: Each subgroup's distribution is generated by integrating the item response curve over the authentic demographic trait distribution:
$$P(Y_{ij} \leq k \mid g) = \Phi\left(\frac{\tau_{j,k} - \alpha_j \mu_g}{\sqrt{\alpha_j^2 \sigma_g^2 + 1}}\right)$$
where α_j is either debiased directly from the LLM ($\alpha_j = \kappa \cdot \alpha_j^{LLM}$ with $\kappa \approx 3.0$) or drawn from psychometric reference items.

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4. Empirical Benchmark Scoreboard & Item Breakdown

Table 2 compares the benchmark performance of all paradigms scored against the full CES 2025 ground truth ($N = 20,180$).

Table 2: Benchmark Scoreboard (Mean Total Variation Distance)

Approach / Paradigm	Model / Configuration	Subgroup Cell TV	National TV	Notes
Human Floor	Split-halves of CES 2025	**0.062**	**0.062**	Measurement error lower bound
Cadieux Empirical Bound	Real 2025 National + 2021 Survey Gaps	**0.064**	**0.000**	Perfect macro + historical survey gaps
National Reference	Real 2025 National (flat, zero gaps)	**0.085**	**0.000**	Flat national distribution for all groups
IRT Oracle Ceiling	**Real 2025 National + IRT Latent Gaps**	**0.098**	**0.016**	**IRT upper bound**: Perfect macro + IRT gaps
Historical Survey	**CES 2021 Same Subgroup (Cadieux)**	**0.151**	**0.141**	2021 poll unchanged (beats raw LLMs)
Statistical Baseline	**CES 2021 Multinomial Logit (Savoie)**	**0.175**	**0.150**	Classical demographic regression
Cadieux (Fine-Tuned)	**Llama 70B: Fixed context, 50k**	**0.178**	**0.157**	**Hubert's best fine-tuned model**
Cadieux (Fine-Tuned)	Llama 70B: Semantic context, 50k	0.188	0.165	Semantic nearest-neighbor context
Savoie (IRT Debiased)	**Qwen 9B Partisan + IRT Filter**	**0.291**	**0.272**	**IRT debiased α on prompted run**
Savoie (Raw Prompted)	**Qwen 9B: Rich + Partisan**	**0.302**	**0.258**	**Justin's best raw prompted run**
Untrained Baseline	Untrained Llama-3.3-70B (Cadieux)	0.325	0.296	Raw pretraining prior
Savoie (IRT Debiased)	**Qwen 9B Basic + IRT Filter**	**0.379**	**0.376**	**IRT debiased α on basic run**
Savoie (Raw Prompted)	**Qwen 9B: Basic Demographics**	**0.385**	**0.357**	Region, age, gender only
Savoie (Raw Prompted)	Qwen 2.4T MoE: Basic	0.520	0.477	Severe mode collapse on neutral options

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Table 3: Item-by-Item Comparison (Subgroup Cell TV)

Question	Hubert (Fine-Tuned 70B)	Justin (Prompted 9B)	Raw Justin + IRT Debiasing	IRT Oracle (Real Nat'l + IRT)	CES 2021 Poll	Substantive Dynamic
Affordable Housing	0.10	0.128	**0.100**	**0.061**	0.08	**IRT closes gap**: matches Hubert exactly
MAID (Assisted Death)	0.14	0.228	**0.164**	**0.111**	0.09	**IRT cuts error by 6.4 points**
Jobs vs. Environment	0.15	0.305	**0.261**	**0.096**	0.13	**IRT cuts error by 4.5 points**
Democracy Satisfaction	0.12	0.234	**0.215**	**0.088**	0.06	**IRT improves on raw LLM**

Question	Hubert (Fine-Tuned 70B)	Justin Raw (Prompted 9B)	Justin + IRT Debiasing	IRT Oracle (Real Nat'l + IRT)	CES 2021 Poll	Substantive Dynamic
Personal Finances	0.21	0.213	**0.202**	**0.104**	0.13	**IRT slightly edges out both models**
Pipelines (Energy)	0.17	0.313	0.317	**0.140**	0.16	Prompted LLM over-polarized Prairies vs QC
Spending: Defence	**0.17**	**0.243**	0.285	**0.101**	0.31	Models beat poll; captures hawkish shift
National Childcare	0.18	0.313	0.330	**0.090**	0.09	Prompted LLM underestimates national support
Reconciliation	0.24	0.441	0.456	**0.122**	0.17	Prompted progressive skew (80%+ "spend more")
Immigration Levels	0.30	0.602	**0.594**	**0.065**	0.31	Prompted LLM refused "fewer"; Azure stripped
Average Subgroup TV	**0.178**	**0.302**	**0.291**	**0.098**	**0.151**	**IRT debiasing improves 6/10 items; Oracle reaches 0.098**

5. Why This Methodological Shift Matters for Novel Questions

Comparing silicon sampling against classical multinomial logits or past polls on **repeated questions** creates an artificial benchmark bias: - **The Classical Failure Mode**: A multinomial logit or historical poll requires historical ground-truth labels (Y). On **novel, newly emerging, or unpolled policy items**, historical labels do not exist. Fitting a logit or taking past survey averages is mathematically impossible. - **The IRT Advantage**: In human public opinion, demographic cleavages along latent dimensions θ (regional alienation, age polarization, economic vs. cultural ideology) generalize across novel policy domains. - **The Operational Synergy**: 1. For novel questions, we query the LLM solely for the **item's polarity (sign (α_j))** and **macro consensus cutoffs ($\vec{\tau}_j$)**. 2. We rescale α_j using the debiasing constant ($\kappa \approx 3.0$) and integrate over historical demographic distributions $p(\theta \mid g)$. 3. This generates authentic subgroup distributions on completely novel questions where classical methods fail entirely.

6. Actionable Joint Next Steps for Hubert & Justin

1. **Benchmark Novel CES 2025 Items**:
   - Extract 10–15 policy questions from CES 2025 that were **never asked in 2019 or 2021** (e.g., artificial intelligence regulation, foreign interference, specific housing tax policies).
   - Evaluate prompted models, fine-tuned models, and the IRT combiner on this zero-prior benchmark.
2. **Unfiltered Fine-Tuning Pipeline**:
   - Re-train `Llama-3.3-70B` on unmoderated infrastructure (RunPod / Together) to bypass Azure Content Safety and re-incorporate immigration and moral values into the training corpus.
3. **Implement the Unified Hybrid Estimator**:
   - Combine **Cadieux's fine-tuned model for macro cutoffs ($\vec{\tau}_j$)** with **Savoie's IRT debiasing filter for demographic dispersion ($\alpha_j\theta$)**.
   - As demonstrated by the Oracle test (**Subgroup TV = 0.098**), combining accurate macro sentiment with IRT demographic dispersion provides the clearest path to outperforming both raw LLMs and historical polls.